lab3-korespond.R

Jamistrz

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```r
pop <- read.csv2("Food_Preference2.csv")
View(pop)
library(MASS)
library(ca)
library(RColorBrewer)
library(vcd)
```

```
## Ładowanie wymaganego pakietu: grid
```

```r
library(cluster)
library(FactoMineR)
library(factoextra)
```

```
## Ładowanie wymaganego pakietu: ggplot2
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```r
library(ade4)
```

```
##
## Dołączanie pakietu: 'ade4'
```

```
## Następujący obiekt został zakryty z 'package:FactoMineR':
##
##     reconst
```

```r
library(ExPosition)
```

```
## Ładowanie wymaganego pakietu: prettyGraphs
```

```r
library(ggplot2)
library(reshape2)
library(homals)
```

```
##
## Dołączanie pakietu: 'homals'
```

```
## Następujący obiekt został zakryty z 'package:MASS':
##
##     mammals
```

```r
X<-data.frame(pop)
head(X)
```

```
##   Nationality             Food             Juice Dessert
## 1      Indian Traditional food       Fresh Juice   Maybe
## 2      Indian     Western Food Carbonated drinks     Yes
## 3      Indian     Western Food       Fresh Juice   Maybe
## 4      Indian Traditional food       Fresh Juice   Maybe
## 5      Indian Traditional food       Fresh Juice   Maybe
## 6      Indian Traditional food       Fresh Juice     Yes
```

```r
# ANALIZA MCA

print("wielowymiarowa analiza korespondencji 4 ostatnie pakiety wgrywane")
```

```
## [1] "wielowymiarowa analiza korespondencji 4 ostatnie pakiety wgrywane"
```

```r
print("wybieramy zmienne do analizy")
```

```
## [1] "wybieramy zmienne do analizy"
```

```r
df1<-data.frame(X[,1:4])
head(df1)
```

```
##   Nationality             Food             Juice Dessert
## 1      Indian Traditional food       Fresh Juice   Maybe
## 2      Indian     Western Food Carbonated drinks     Yes
## 3      Indian     Western Food       Fresh Juice   Maybe
## 4      Indian Traditional food       Fresh Juice   Maybe
## 5      Indian Traditional food       Fresh Juice   Maybe
## 6      Indian Traditional food       Fresh Juice     Yes
```

```r
print("wybieramy aktywne przypadki i zmienne")
```

```
## [1] "wybieramy aktywne przypadki i zmienne"
```

```r
df1.active <- df1[, 1:4]
head(df1.active)
```

```
##   Nationality             Food             Juice Dessert
## 1      Indian Traditional food       Fresh Juice   Maybe
## 2      Indian     Western Food Carbonated drinks     Yes
## 3      Indian     Western Food       Fresh Juice   Maybe
## 4      Indian Traditional food       Fresh Juice   Maybe
## 5      Indian Traditional food       Fresh Juice   Maybe
## 6      Indian Traditional food       Fresh Juice     Yes
```

```r
summary(df1.active)
```

```
##  Nationality            Food              Juice             Dessert
##  Length:273         Length:273         Length:273         Length:273
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
```

```r
cats<- apply(X, 2, function(x) nlevels(as.factor(x)))
print(cats)
```

```
## Nationality        Food       Juice     Dessert
##           4           2           2           3
```

```r
unique(pop[1])
```

```
##    Nationality
## 1       Indian
## 8    Pakistani
## 12  Indonesian
## 15   Malaysian
```

```r
unique(pop[2])
```

```
##               Food
## 1 Traditional food
## 2     Western Food
```

```
unique(pop[3])
```

```
##                Juice
## 1        Fresh Juice
## 2 Carbonated drinks
```

```
unique(pop[4])
```

```
##    Dessert
## 1    Maybe
## 2      Yes
## 15      No
```

```
nationality_counts <- table(pop$Nationality)
nationality_counts
```

```
##
##     Indian Indonesian  Malaysian  Pakistani
##        241         10         16          6
```

```
food_counts <- table(pop$Food)
food_counts
```

```
##
## Traditional food     Western Food
##              228               45
```

```
juice_counts <- table(pop$Juice)
juice_counts
```

```
##
## Carbonated drinks       Fresh Juice
##                30               243
```

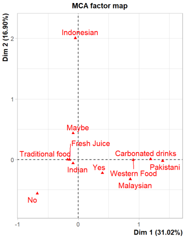
```
dessert_counts <- table(pop$Dessert)
dessert_counts
```

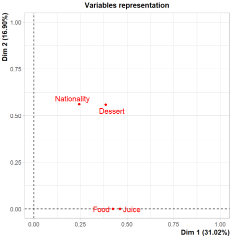
```
##
## Maybe    No   Yes
##   117    48   108
```

```
print("tworzymy obiekt do analizy - pakiet MCA")
```

```
## [1] "tworzymy obiekt do analizy - pakiet MCA"
```

```
res.mca <- MCA(df1.active,ncp = 2, graph = TRUE,method = "Burt")
```





```
#Podsumowanie
summary(res.mca)
```

```
##
## Call:
## MCA(X = df1.active, ncp = 2, graph = TRUE, method = "Burt")
##
##
## Eigenvalues
##                        Dim.1   Dim.2   Dim.3   Dim.4   Dim.5   Dim.6   Dim.7
## Variance               0.144   0.078   0.063   0.056   0.048   0.041   0.033
## % of var.             31.018  16.900  13.695  12.019  10.283   8.889   7.195
## Cumulative % of var.  31.018  47.918  61.613  73.632  83.915  92.805 100.000
##
## Individuals (the 10 first)
##                        Dim.1    ctr   cos2    Dim.2    ctr   cos2  
## 1                 | -0.331  0.106  0.246 |  0.335  0.147  0.251 |
## 2                 |  1.583  2.424  0.676 | -0.252  0.083  0.017 |
## 3                 |  0.382  0.141  0.088 |  0.328  0.141  0.065 |
## 4                 | -0.331  0.106  0.246 |  0.335  0.147  0.251 |
## 5                 | -0.331  0.106  0.246 |  0.335  0.147  0.251 |
## 6                 | -0.012  0.000  0.000 | -0.253  0.084  0.129 |
## 7                 | -0.012  0.000  0.000 | -0.253  0.084  0.129 |
## 8                 |  2.558  6.325  0.442 | -0.216  0.061  0.003 |
## 9                 | -0.012  0.000  0.000 | -0.253  0.084  0.129 |
## 10                |  0.701  0.475  0.287 | -0.260  0.089  0.039 |
##
## Categories (the 10 first)
##                        Dim.1    ctr   cos2 v.test    Dim.2    ctr   cos2 v.test  
## Indian            | -0.089  1.225  0.232 -4.041 | -0.061  1.058  0.109 -2.772 |
## Indonesian        | -0.050  0.016  0.000 -0.161 |  2.006 47.076  0.603  6.450 |
## Malaysian         |  0.856  7.467  0.177  3.521 | -0.323  1.956  0.025 -1.330 |
## Pakistani         |  1.388  7.370  0.169  3.431 | -0.021  0.003  0.000 -0.051 |
## Traditional food  | -0.178  4.618  0.580 -6.616 |  0.001  0.000  0.000  0.047 |
## Western Food      |  0.903 23.397  0.580  6.616 | -0.006  0.002  0.000 -0.047 |
## Carbonated drinks |  1.190 27.110  0.624  6.899 |  0.008  0.002  0.000  0.045 |
## Fresh Juice       | -0.147  3.347  0.624 -6.899 | -0.001  0.000  0.000 -0.045 |
## Maybe             | -0.088  0.572  0.023 -1.251 |  0.435 25.967  0.559  6.220 |
## No                | -0.678 14.069  0.367 -5.165 | -0.561 17.682  0.251 -4.274 |
##
## Categorical variables (eta2)
##                       Dim.1 Dim.2  
## Nationality       | 0.244 0.561 |
## Food              | 0.425 0.000 |
## Juice             | 0.462 0.000 |
## Dessert           | 0.386 0.558 |
```

```
"wartosci wlasne"
```

```
## [1] "wartosci wlasne"
```

```
res.mca$eig
```

```
##       eigenvalue percentage of variance cumulative percentage of variance
## dim 1 0.14361382              31.017903                          31.01790
## dim 2 0.07824966              16.900466                          47.91837
## dim 3 0.06340842              13.695034                          61.61340
## dim 4 0.05564749              12.018818                          73.63222
## dim 5 0.04761024              10.282923                          83.91514
## dim 6 0.04115819               8.889400                          92.80454
## dim 7 0.03331518               7.195457                         100.00000
```

```
print(res.mca)
```

```
## **Results of the Multiple Correspondence Analysis (MCA)**
## The analysis was performed on 273 individuals, described by 4 variables
## *The results are available in the following objects:
##
##    name              description                       
## 1  "$eig"            "eigenvalues"                     
## 2  "$var"            "results for the variables"       
## 3  "$var$coord"      "coord. of the categories"        
## 4  "$var$cos2"       "cos2 for the categories"         
## 5  "$var$contrib"    "contributions of the categories" 
## 6  "$var$v.test"     "v-test for the categories"       
## 7  "$var$eta2"       "coord. of variables"             
## 8  "$ind"            "results for the individuals"     
## 9  "$ind$coord"      "coord. for the individuals"      
## 10 "$ind$cos2"       "cos2 for the individuals"        
## 11 "$ind$contrib"    "contributions of the individuals"
## 12 "$call"           "intermediate results"            
## 13 "$call$marge.col" "weights of columns"              
## 14 "$call$marge.li"  "weights of rows"
```

```
res.mca$ind$coord
```

```
##            Dim 1       Dim 2
## 1   -0.331188005  0.33473056
## 2    1.583498560 -0.25228675
## 3    0.382086424  0.32792685
## 4   -0.331188005  0.33473056
## 5   -0.331188005  0.33473056
## 6   -0.012068002 -0.25325281
## 7   -0.012068002 -0.25325281
## 8    2.557982565 -0.21604179
## 9   -0.012068002 -0.25325281
## 10   0.701206427 -0.26005652
## 11  -0.012068002 -0.25325281
## 12  -0.305241965  2.18192644
## 13   1.583498560 -0.25228675
## 14   0.643296000  0.37097553
## 15  -0.097395149 -0.79001887
## 16   0.292098548  0.10059374
## 17  -0.331188005  0.33473056
## 18   1.174390681  0.10836350
## 19  -0.305241965  2.18192644
## 20   1.005372978  0.09379003
## 21   1.324492981 -0.49419335
## 22   0.611218551 -0.48738964
## 23   0.013878038  1.59394307
## 24  -0.331188005  0.33473056
## 25  -0.012068002 -0.25325281
## 26  -0.012068002 -0.25325281
## 27   1.174390681  0.10836350
## 28   1.324492981 -0.49419335
## 29   0.382086424  0.32792685
## 30   0.382086424  0.32792685
## 31   0.382086424  0.32792685
## 32   1.324492981 -0.49419335
## 33   0.382086424  0.32792685
## 34  -0.012068002 -0.25325281
## 35   0.292098548  0.10059374
## 36  -0.331188005  0.33473056
## 37   0.643296000  0.37097553
## 38   0.962416003 -0.21700784
## 39   1.844708135 -0.20923808
## 40   2.206785113 -0.48642358
## 41   1.675690433 -0.22381155
## 42   0.701206427 -0.26005652
## 43  -0.331188005  0.33473056
## 44  -0.331188005  0.33473056
## 45  -0.720681703 -0.55588204
## 46  -0.012068002 -0.25325281
## 47  -0.331188005  0.33473056
## 48   1.583498560 -0.25228675
## 49  -0.012068002 -0.25325281
## 50  -0.331188005  0.33473056
## 51  -0.720681703 -0.55588204
## 52   1.264378557  0.33569662
## 53  -0.331188005  0.33473056
```

```
## 54   0.382086424  0.32792685
## 55   0.870224130 -0.24548304
## 56   0.551104127  0.34250033
## 57  -0.012068002 -0.25325281
## 58  -0.012068002 -0.25325281
## 59  -0.305241965  2.18192644
## 60   1.609444599  1.59490913
## 61  -0.012068002 -0.25325281
## 62  -0.720681703 -0.55588204
## 63  -0.331188005  0.33473056
## 64   0.013878038  1.59394307
## 65  -0.012068002 -0.25325281
## 66   0.870224130 -0.24548304
## 67  -0.012068002 -0.25325281
## 68  -0.331188005  0.33473056
## 69   0.611218551 -0.48738964
## 70  -0.012068002 -0.25325281
## 71  -0.720681703 -0.55588204
## 72  -0.720681703 -0.55588204
## 73   0.611218551 -0.48738964
## 74  -0.012068002 -0.25325281
## 75  -0.720681703 -0.55588204
## 76  -0.331188005  0.33473056
## 77  -0.012068002 -0.25325281
## 78  -0.720681703 -0.55588204
## 79   0.611218551 -0.48738964
## 80  -0.305241965  2.18192644
## 81  -0.720681703 -0.55588204
## 82   0.611218551 -0.48738964
## 83  -0.007407273 -0.56268575
## 84  -0.305241965  2.18192644
## 85   1.583498560 -0.25228675
## 86   1.264378557  0.33569662
## 87   0.870224130 -0.24548304
## 88  -0.331188005  0.33473056
## 89  -0.012068002 -0.25325281
## 90   0.611218551 -0.48738964
## 91   0.382086424  0.32792685
## 92  -0.012068002 -0.25325281
## 93  -0.012068002 -0.25325281
## 94  -0.012068002 -0.25325281
## 95  -0.012068002 -0.25325281
## 96   1.583498560 -0.25228675
## 97   0.701206427 -0.26005652
## 98   0.701206427 -0.26005652
## 99  -0.720681703 -0.55588204
## 100 -0.720681703 -0.55588204
## 101 -0.720681703 -0.55588204
## 102 -0.012068002 -0.25325281
## 103 -0.331188005  0.33473056
## 104 -0.012068002 -0.25325281
## 105 -0.720681703 -0.55588204
## 106 -0.012068002 -0.25325281
## 107 -0.331188005  0.33473056
## 108 -0.720681703 -0.55588204
## 109 -0.007407273 -0.56268575
## 110 -0.331188005  0.33473056
## 111  0.870224130 -0.24548304
## 112  0.551104127  0.34250033
## 113 -0.305241965  2.18192644
## 114 -0.305241965  2.18192644
## 115 -0.331188005  0.33473056
## 116 -0.331188005  0.33473056
## 117 -0.331188005  0.33473056
## 118 -0.720681703 -0.55588204
## 119 -0.720681703 -0.55588204
## 120 -0.331188005  0.33473056
## 121 -0.012068002 -0.25325281
## 122 -0.331188005  0.33473056
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## 126 -0.331188005  0.33473056
## 127  1.583498560 -0.25228675
## 128 -0.012068002 -0.25325281
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## 140  1.583498560 -0.25228675
## 141 -0.331188005  0.33473056
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## 143 -0.331188005  0.33473056
## 144  0.551104127  0.34250033
## 145 -0.331188005  0.33473056
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## 148 -0.331188005  0.33473056
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## 166 -0.331188005  0.33473056
## 167 -0.331188005  0.33473056
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## 169 -0.331188005  0.33473056
## 170  0.701206427 -0.26005652
## 171 -0.331188005  0.33473056
## 172 -0.012068002 -0.25325281
## 173 -0.331188005  0.33473056
## 174 -0.012068002 -0.25325281
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## 176 -0.012068002 -0.25325281
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## 180 -0.331188005  0.33473056
## 181 -0.720681703 -0.55588204
## 182 -0.720681703 -0.55588204
## 183 -0.720681703 -0.55588204
## 184 -0.331188005  0.33473056
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## 186 -0.331188005  0.33473056
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## 190 -0.331188005  0.33473056
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## 194 -0.331188005  0.33473056
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## 196 -0.720681703 -0.55588204
```

```
## 197 -0.012068002 -0.25325281
## 198 -0.720681703 -0.55588204
## 199  0.551104127  0.34250033
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## 202  0.382086424  0.32792685
## 203 -0.720681703 -0.55588204
## 204 -0.331188005  0.33473056
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## 215  0.382086424  0.32792685
## 216 -0.331188005  0.33473056
## 217  0.551104127  0.34250033
## 218  0.701206427 -0.26005652
## 219  0.382086424  0.32792685
## 220 -0.720681703 -0.55588204
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## 231 -0.012068002 -0.25325281
## 232 -0.720681703 -0.55588204
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## 234 -0.331188005  0.33473056
## 235 -0.012068002 -0.25325281
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## 237 -0.012068002 -0.25325281
## 238 -0.331188005  0.33473056
## 239 -0.331188005  0.33473056
## 240 -0.720681703 -0.55588204
## 241 -0.331188005  0.33473056
## 242 -0.012068002 -0.25325281
## 243 -0.012068002 -0.25325281
## 244 -0.720681703 -0.55588204
## 245 -0.012068002 -0.25325281
## 246 -0.720681703 -0.55588204
## 247 -0.331188005  0.33473056
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## 250 -0.012068002 -0.25325281
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## 253 -0.331188005  0.33473056
## 254 -0.012068002 -0.25325281
## 255 -0.331188005  0.33473056
## 256  0.870224130 -0.24548304
## 257 -0.331188005  0.33473056
## 258 -0.331188005  0.33473056
## 259 -0.331188005  0.33473056
## 260  0.382086424  0.32792685
## 261 -0.331188005  0.33473056
## 262  0.870224130 -0.24548304
## 263 -0.012068002 -0.25325281
## 264  0.870224130 -0.24548304
## 265 -0.331188005  0.33473056
## 266 -0.331188005  0.33473056
## 267 -0.331188005  0.33473056
## 268 -0.720681703 -0.55588204
## 269  0.701206427 -0.26005652
## 270 -0.012068002 -0.25325281
## 271 -0.012068002 -0.25325281
## 272 -0.012068002 -0.25325281
## 273 -0.720681703 -0.55588204
```

```
print("teraz mozemy obserwowac rozne wartosci dla multiple correspondense analysis")
```

```
## [1] "teraz mozemy obserwowac rozne wartosci dla multiple correspondense analysis"
```

```
print("np. eigen value")
```

```
## [1] "np. eigen value"
```

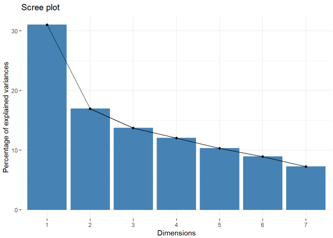
```
eig.val <- get_eigenvalue(res.mca)
head(eig.val)
```

```
##       eigenvalue variance.percent cumulative.variance.percent
## Dim.1 0.14361382         31.01790                    31.01790
## Dim.2 0.07824966         16.90047                    47.91837
## Dim.3 0.06340842         13.69503                    61.61340
## Dim.4 0.05564749         12.01882                    73.63222
## Dim.5 0.04761024         10.28292                    83.91514
## Dim.6 0.04115819          8.88940                    92.80454
```

```
print("wykres eigen value")
```

```
## [1] "wykres eigen value"
```

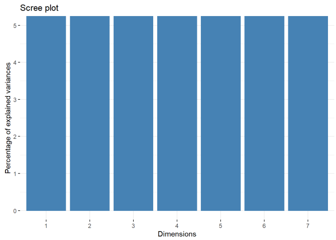
```
fviz_eig(res.mca)
```



```
print("jak spadaja warto?sci")
```

```
## [1] "jak spadaja warto?sci"
```

```
fviz_screeplot(res.mca, addlabels = TRUE, ylim = c(0, 5))
```



```
print("przegl?d zmiennosci i zale?no?ci")
```

```
## [1] "przegl?d zmiennosci i zale?no?ci"
```

```
var <- get_mca_var(res.mca)
"opis co dostajemy var$..."
```

```
## [1] "opis co dostajemy var$..."
```

```
print(var)
```

```
## Multiple Correspondence Analysis Results for variables
##  ===================================================
##   Name       Description
## 1 "$coord"   "Coordinates for categories"
## 2 "$cos2"    "Cos2 for categories"
## 3 "$contrib" "contributions of categories"
```

```
print("warto?ci coordynant")
```

```
## [1] "warto?ci coordynant"
```

```
print(var$coord)
```

```
##                         Dim 1         Dim 2
## Indian            -0.08927978 -0.0612467897
## Indonesian        -0.04994931  2.0056280365
## Malaysian          0.85553323 -0.3232284899
## Pakistani          1.38789819 -0.0206913668
## Traditional food  -0.17822336  0.0012548646
## Western Food       0.90299836 -0.0063579806
## Carbonated drinks  1.19045817  0.0077384301
## Fresh Juice       -0.14697014 -0.0009553617
## Maybe             -0.08756022  0.4354858967
## No                -0.67797680 -0.5610433751
## Yes                0.39617993 -0.2224237769
```

```
print("warto?ci cos^2 - korelacji z nowymi zmiennymi")
```

```
## [1] "warto?ci cos^2 - korelacji z nowymi zmiennymi"
```

```
print(var$cos2)
```

```
##                          Dim 1        Dim 2
## Indian            0.2315201118 1.089555e-01
## Indonesian        0.0003737396 6.025748e-01
## Malaysian         0.1772248553 2.529706e-02
## Pakistani         0.1689725976 3.755601e-05
## Traditional food  0.5803325586 2.877010e-05
## Western Food      0.5803325586 2.877010e-05
## Carbonated drinks 0.6243504199 2.638189e-05
## Fresh Juice       0.6243504199 2.638189e-05
## Maybe             0.0226153027 5.594186e-01
## No                0.3666489340 2.510809e-01
## Yes               0.3851092045 1.213840e-01
```

```
print("dla indywidualnych")
```

```
## [1] "dla indywidualnych"
```

```
ind <- get_mca_ind(res.mca)
head(ind)
```

```
## $coord
##             Dim 1       Dim 2
## 1   -0.331188005  0.33473056
## 2    1.583498560 -0.25228675
## 3    0.382086424  0.32792685
## 4   -0.331188005  0.33473056
## 5   -0.331188005  0.33473056
## 6   -0.012068002 -0.25325281
## 7   -0.012068002 -0.25325281
## 8    2.557982565 -0.21604179
## 9   -0.012068002 -0.25325281
## 10   0.701206427 -0.26005652
## 11  -0.012068002 -0.25325281
## 12  -0.305241965  2.18192644
## 13   1.583498560 -0.25228675
## 14   0.643296000  0.37097553
## 15  -0.097395149 -0.79001887
## 16   0.292098548  0.10059374
## 17  -0.331188005  0.33473056
## 18   1.174390681  0.10836350
## 19  -0.305241965  2.18192644
## 20   1.005372978  0.09379003
## 21   1.324492981 -0.49419335
## 22   0.611218551 -0.48738964
## 23   0.013878038  1.59394307
## 24  -0.331188005  0.33473056
## 25  -0.012068002 -0.25325281
## 26  -0.012068002 -0.25325281
## 27   1.174390681  0.10836350
## 28   1.324492981 -0.49419335
## 29   0.382086424  0.32792685
## 30   0.382086424  0.32792685
## 31   0.382086424  0.32792685
## 32   1.324492981 -0.49419335
## 33   0.382086424  0.32792685
## 34  -0.012068002 -0.25325281
## 35   0.292098548  0.10059374
## 36  -0.331188005  0.33473056
## 37   0.643296000  0.37097553
## 38   0.962416003 -0.21700784
## 39   1.844708135 -0.20923808
```

```
## 40   2.206785113 -0.48642358
## 41   1.675690433 -0.22381155
## 42   0.701206427 -0.26005652
## 43  -0.331188005  0.33473056
## 44  -0.331188005  0.33473056
## 45  -0.720681703 -0.55588204
## 46  -0.012068002 -0.25325281
## 47  -0.331188005  0.33473056
## 48   1.583498560 -0.25228675
## 49  -0.012068002 -0.25325281
## 50  -0.331188005  0.33473056
## 51  -0.720681703 -0.55588204
## 52   1.264378557  0.33569662
## 53  -0.331188005  0.33473056
## 54   0.382086424  0.32792685
## 55   0.870224130 -0.24548304
## 56   0.551104127  0.34250033
## 57  -0.012068002 -0.25325281
## 58  -0.012068002 -0.25325281
## 59  -0.305241965  2.18192644
## 60   1.609444599  1.59490913
## 61  -0.012068002 -0.25325281
## 62  -0.720681703 -0.55588204
## 63  -0.331188005  0.33473056
## 64   0.013878038  1.59394307
## 65  -0.012068002 -0.25325281
## 66   0.870224130 -0.24548304
## 67  -0.012068002 -0.25325281
## 68  -0.331188005  0.33473056
## 69   0.611218551 -0.48738964
## 70  -0.012068002 -0.25325281
## 71  -0.720681703 -0.55588204
## 72  -0.720681703 -0.55588204
## 73   0.611218551 -0.48738964
## 74  -0.012068002 -0.25325281
## 75  -0.720681703 -0.55588204
## 76  -0.331188005  0.33473056
## 77  -0.012068002 -0.25325281
## 78  -0.720681703 -0.55588204
## 79   0.611218551 -0.48738964
## 80  -0.305241965  2.18192644
## 81  -0.720681703 -0.55588204
## 82   0.611218551 -0.48738964
## 83  -0.007407273 -0.56268575
## 84  -0.305241965  2.18192644
## 85   1.583498560 -0.25228675
## 86   1.264378557  0.33569662
## 87   0.870224130 -0.24548304
## 88  -0.331188005  0.33473056
## 89  -0.012068002 -0.25325281
## 90   0.611218551 -0.48738964
## 91   0.382086424  0.32792685
## 92  -0.012068002 -0.25325281
## 93  -0.012068002 -0.25325281
## 94  -0.012068002 -0.25325281
## 95  -0.012068002 -0.25325281
## 96   1.583498560 -0.25228675
## 97   0.701206427 -0.26005652
## 98   0.701206427 -0.26005652
## 99  -0.720681703 -0.55588204
## 100 -0.720681703 -0.55588204
## 101 -0.720681703 -0.55588204
## 102 -0.012068002 -0.25325281
## 103 -0.331188005  0.33473056
## 104 -0.012068002 -0.25325281
## 105 -0.720681703 -0.55588204
## 106 -0.012068002 -0.25325281
## 107 -0.331188005  0.33473056
## 108 -0.720681703 -0.55588204
## 109 -0.007407273 -0.56268575
## 110 -0.331188005  0.33473056
## 111  0.870224130 -0.24548304
## 112  0.551104127  0.34250033
## 113 -0.305241965  2.18192644
## 114 -0.305241965  2.18192644
## 115 -0.331188005  0.33473056
## 116 -0.331188005  0.33473056
## 117 -0.331188005  0.33473056
## 118 -0.720681703 -0.55588204
## 119 -0.720681703 -0.55588204
## 120 -0.331188005  0.33473056
## 121 -0.012068002 -0.25325281
## 122 -0.331188005  0.33473056
## 123 -0.331188005  0.33473056
## 124 -0.720681703 -0.55588204
## 125 -0.720681703 -0.55588204
## 126 -0.331188005  0.33473056
## 127  1.583498560 -0.25228675
## 128 -0.012068002 -0.25325281
## 129 -0.720681703 -0.55588204
## 130 -0.012068002 -0.25325281
## 131  0.382086424  0.32792685
## 132 -0.012068002 -0.25325281
## 133 -0.331188005  0.33473056
## 134 -0.331188005  0.33473056
## 135 -0.331188005  0.33473056
## 136  0.701206427 -0.26005652
## 137 -0.012068002 -0.25325281
## 138 -0.012068002 -0.25325281
## 139 -0.012068002 -0.25325281
## 140  1.583498560 -0.25228675
## 141 -0.331188005  0.33473056
## 142 -0.012068002 -0.25325281
## 143 -0.331188005  0.33473056
## 144  0.551104127  0.34250033
## 145 -0.331188005  0.33473056
## 146  0.382086424  0.32792685
## 147 -0.720681703 -0.55588204
## 148 -0.331188005  0.33473056
## 149 -0.331188005  0.33473056
## 150 -0.331188005  0.33473056
## 151 -0.331188005  0.33473056
## 152 -0.012068002 -0.25325281
## 153 -0.720681703 -0.55588204
## 154 -0.012068002 -0.25325281
## 155  0.870224130 -0.24548304
## 156 -0.720681703 -0.55588204
## 157 -0.012068002 -0.25325281
## 158  1.264378557  0.33569662
## 159 -0.012068002 -0.25325281
## 160 -0.720681703 -0.55588204
## 161  0.551104127  0.34250033
## 162 -0.331188005  0.33473056
## 163  0.701206427 -0.26005652
## 164 -0.331188005  0.33473056
## 165 -0.720681703 -0.55588204
## 166 -0.331188005  0.33473056
## 167 -0.331188005  0.33473056
## 168 -0.331188005  0.33473056
## 169 -0.331188005  0.33473056
## 170  0.701206427 -0.26005652
## 171 -0.331188005  0.33473056
## 172 -0.012068002 -0.25325281
## 173 -0.331188005  0.33473056
## 174 -0.012068002 -0.25325281
## 175 -0.720681703 -0.55588204
## 176 -0.012068002 -0.25325281
## 177 -0.012068002 -0.25325281
## 178 -0.331188005  0.33473056
## 179 -0.331188005  0.33473056
## 180 -0.331188005  0.33473056
## 181 -0.720681703 -0.55588204
## 182 -0.720681703 -0.55588204
```

```
## 183 -0.720681703 -0.55588204
## 184 -0.331188005  0.33473056
## 185 -0.331188005  0.33473056
## 186 -0.331188005  0.33473056
## 187 -0.012068002 -0.25325281
## 188 -0.720681703 -0.55588204
## 189 -0.720681703 -0.55588204
## 190 -0.331188005  0.33473056
## 191 -0.331188005  0.33473056
## 192 -0.331188005  0.33473056
## 193 -0.720681703 -0.55588204
## 194 -0.331188005  0.33473056
## 195 -0.720681703 -0.55588204
## 196 -0.720681703 -0.55588204
## 197 -0.012068002 -0.25325281
## 198 -0.720681703 -0.55588204
## 199  0.551104127  0.34250033
## 200 -0.331188005  0.33473056
## 201  0.382086424  0.32792685
## 202  0.382086424  0.32792685
## 203 -0.720681703 -0.55588204
## 204 -0.331188005  0.33473056
## 205 -0.012068002 -0.25325281
## 206 -0.331188005  0.33473056
## 207 -0.720681703 -0.55588204
## 208 -0.331188005  0.33473056
## 209 -0.331188005  0.33473056
## 210 -0.012068002 -0.25325281
## 211 -0.012068002 -0.25325281
## 212 -0.331188005  0.33473056
## 213 -0.012068002 -0.25325281
## 214 -0.720681703 -0.55588204
## 215  0.382086424  0.32792685
## 216 -0.331188005  0.33473056
## 217  0.551104127  0.34250033
## 218  0.701206427 -0.26005652
## 219  0.382086424  0.32792685
## 220 -0.720681703 -0.55588204
## 221 -0.331188005  0.33473056
## 222 -0.012068002 -0.25325281
## 223 -0.012068002 -0.25325281
## 224 -0.331188005  0.33473056
## 225 -0.331188005  0.33473056
## 226 -0.012068002 -0.25325281
## 227 -0.012068002 -0.25325281
## 228  0.382086424  0.32792685
## 229 -0.012068002 -0.25325281
## 230 -0.331188005  0.33473056
## 231 -0.012068002 -0.25325281
## 232 -0.720681703 -0.55588204
## 233 -0.720681703 -0.55588204
## 234 -0.331188005  0.33473056
## 235 -0.012068002 -0.25325281
## 236 -0.331188005  0.33473056
## 237 -0.012068002 -0.25325281
## 238 -0.331188005  0.33473056
## 239 -0.331188005  0.33473056
## 240 -0.720681703 -0.55588204
## 241 -0.331188005  0.33473056
## 242 -0.012068002 -0.25325281
## 243 -0.012068002 -0.25325281
## 244 -0.720681703 -0.55588204
## 245 -0.012068002 -0.25325281
## 246 -0.720681703 -0.55588204
## 247 -0.331188005  0.33473056
## 248 -0.012068002 -0.25325281
## 249 -0.720681703 -0.55588204
## 250 -0.012068002 -0.25325281
## 251  0.701206427 -0.26005652
## 252 -0.012068002 -0.25325281
## 253 -0.331188005  0.33473056
## 254 -0.012068002 -0.25325281
## 255 -0.331188005  0.33473056
## 256  0.870224130 -0.24548304
## 257 -0.331188005  0.33473056
## 258 -0.331188005  0.33473056
## 259 -0.331188005  0.33473056
## 260  0.382086424  0.32792685
## 261 -0.331188005  0.33473056
## 262  0.870224130 -0.24548304
## 263 -0.012068002 -0.25325281
## 264  0.870224130 -0.24548304
## 265 -0.331188005  0.33473056
## 266 -0.331188005  0.33473056
## 267 -0.331188005  0.33473056
## 268 -0.720681703 -0.55588204
## 269  0.701206427 -0.26005652
## 270 -0.012068002 -0.25325281
## 271 -0.012068002 -0.25325281
## 272 -0.012068002 -0.25325281
## 273 -0.720681703 -0.55588204
## 
## $contrib
##            Dim 1      Dim 2
## 1   1.060202e-01 0.14671917
## 2   2.423676e+00 0.08334605
## 3   1.411115e-01 0.14081538
## 4   1.060202e-01 0.14671917
## 5   1.060202e-01 0.14671917
## 6   1.407700e-04 0.08398557
## 7   1.407700e-04 0.08398557
## 8   6.324619e+00 0.06111835
## 9   1.407700e-04 0.08398557
## 10  4.752597e-01 0.08855878
## 11  1.407700e-04 0.08398557
## 12  9.005912e-02 6.23413684
## 13  2.423676e+00 0.08334605
## 14  4.000009e-01 0.18021322
## 15  9.168829e-03 0.81728032
## 16  8.247039e-02 0.01325067
## 17  1.060202e-01 0.14671917
## 18  1.333105e+00 0.01537667
## 19  9.005912e-02 6.23413684
## 20  9.769980e-01 0.01151886
## 21  1.695659e+00 0.31980843
## 22  3.611040e-01 0.31106325
## 23  1.861639e-04 3.32691519
## 24  1.060202e-01 0.14671917
## 25  1.407700e-04 0.08398557
## 26  1.407700e-04 0.08398557
## 27  1.333105e+00 0.01537667
## 28  1.695659e+00 0.31980843
## 29  1.411115e-01 0.14081538
## 30  1.411115e-01 0.14081538
## 31  1.411115e-01 0.14081538
## 32  1.695659e+00 0.31980843
## 33  1.411115e-01 0.14081538
## 34  1.407700e-04 0.08398557
## 35  8.247039e-02 0.01325067
## 36  1.060202e-01 0.14671917
## 37  4.000009e-01 0.18021322
## 38  8.952924e-01 0.06166617
## 39  3.289232e+00 0.05732942
## 40  4.707164e+00 0.30983135
## 41  2.714106e+00 0.06559355
## 42  4.752597e-01 0.08855878
## 43  1.060202e-01 0.14671917
## 44  1.060202e-01 0.14671917
## 45  5.020260e-01 0.40463309
## 46  1.407700e-04 0.08398557
## 47  1.060202e-01 0.14671917
## 48  2.423676e+00 0.08334605
## 49  1.407700e-04 0.08398557
```

```
## 50  1.060202e-01 0.14671917
## 51  5.020260e-01 0.40463309
## 52  1.545231e+00 0.14756728
## 53  1.060202e-01 0.14671917
## 54  1.411115e-01 0.14081538
## 55  7.319838e-01 0.07891129
## 56  2.935665e-01 0.15360951
## 57  1.407700e-04 0.08398557
## 58  1.407700e-04 0.08398557
## 59  9.005912e-02 6.23413684
## 60  2.503752e+00 3.33094917
## 61  1.407700e-04 0.08398557
## 62  5.020260e-01 0.40463309
## 63  1.060202e-01 0.14671917
## 64  1.861639e-04 3.32691519
## 65  1.407700e-04 0.08398557
## 66  7.319838e-01 0.07891129
## 67  1.407700e-04 0.08398557
## 68  1.060202e-01 0.14671917
## 69  3.611040e-01 0.31106325
## 70  1.407700e-04 0.08398557
## 71  5.020260e-01 0.40463309
## 72  5.020260e-01 0.40463309
## 73  3.611040e-01 0.31106325
## 74  1.407700e-04 0.08398557
## 75  5.020260e-01 0.40463309
## 76  1.060202e-01 0.14671917
## 77  1.407700e-04 0.08398557
## 78  5.020260e-01 0.40463309
## 79  3.611040e-01 0.31106325
## 80  9.005912e-02 6.23413684
## 81  5.020260e-01 0.40463309
## 82  3.611040e-01 0.31106325
## 83  5.303419e-05 0.41459871
## 84  9.005912e-02 6.23413684
## 85  2.423676e+00 0.08334605
## 86  1.545231e+00 0.14756728
## 87  7.319838e-01 0.07891129
## 88  1.060202e-01 0.14671917
## 89  1.407700e-04 0.08398557
## 90  3.611040e-01 0.31106325
## 91  1.411115e-01 0.14081538
## 92  1.407700e-04 0.08398557
## 93  1.407700e-04 0.08398557
## 94  1.407700e-04 0.08398557
## 95  1.407700e-04 0.08398557
## 96  2.423676e+00 0.08334605
## 97  4.752597e-01 0.08855878
## 98  4.752597e-01 0.08855878
## 99  5.020260e-01 0.40463309
## 100 5.020260e-01 0.40463309
## 101 5.020260e-01 0.40463309
## 102 1.407700e-04 0.08398557
## 103 1.060202e-01 0.14671917
## 104 1.407700e-04 0.08398557
## 105 5.020260e-01 0.40463309
## 106 1.407700e-04 0.08398557
## 107 1.060202e-01 0.14671917
## 108 5.020260e-01 0.40463309
## 109 5.303419e-05 0.41459871
## 110 1.060202e-01 0.14671917
## 111 7.319838e-01 0.07891129
## 112 2.935665e-01 0.15360951
## 113 9.005912e-02 6.23413684
## 114 9.005912e-02 6.23413684
## 115 1.060202e-01 0.14671917
## 116 1.060202e-01 0.14671917
## 117 1.060202e-01 0.14671917
## 118 5.020260e-01 0.40463309
## 119 5.020260e-01 0.40463309
## 120 1.060202e-01 0.14671917
## 121 1.407700e-04 0.08398557
## 122 1.060202e-01 0.14671917
## 123 1.060202e-01 0.14671917
## 124 5.020260e-01 0.40463309
## 125 5.020260e-01 0.40463309
## 126 1.060202e-01 0.14671917
## 127 2.423676e+00 0.08334605
## 128 1.407700e-04 0.08398557
## 129 5.020260e-01 0.40463309
## 130 1.407700e-04 0.08398557
## 131 1.411115e-01 0.14081538
## 132 1.407700e-04 0.08398557
## 133 1.060202e-01 0.14671917
## 134 1.060202e-01 0.14671917
## 135 1.060202e-01 0.14671917
## 136 4.752597e-01 0.08855878
## 137 1.407700e-04 0.08398557
## 138 1.407700e-04 0.08398557
## 139 1.407700e-04 0.08398557
## 140 2.423676e+00 0.08334605
## 141 1.060202e-01 0.14671917
## 142 1.407700e-04 0.08398557
## 143 1.060202e-01 0.14671917
## 144 2.935665e-01 0.15360951
## 145 1.060202e-01 0.14671917
## 146 1.411115e-01 0.14081538
## 147 5.020260e-01 0.40463309
## 148 1.060202e-01 0.14671917
## 149 1.060202e-01 0.14671917
## 150 1.060202e-01 0.14671917
## 151 1.060202e-01 0.14671917
## 152 1.407700e-04 0.08398557
## 153 5.020260e-01 0.40463309
## 154 1.407700e-04 0.08398557
## 155 7.319838e-01 0.07891129
## 156 5.020260e-01 0.40463309
## 157 1.407700e-04 0.08398557
## 158 1.545231e+00 0.14756728
## 159 1.407700e-04 0.08398557
## 160 5.020260e-01 0.40463309
## 161 2.935665e-01 0.15360951
## 162 1.060202e-01 0.14671917
## 163 4.752597e-01 0.08855878
## 164 1.060202e-01 0.14671917
## 165 5.020260e-01 0.40463309
## 166 1.060202e-01 0.14671917
## 167 1.060202e-01 0.14671917
## 168 1.060202e-01 0.14671917
## 169 1.060202e-01 0.14671917
## 170 4.752597e-01 0.08855878
## 171 1.060202e-01 0.14671917
## 172 1.407700e-04 0.08398557
## 173 1.060202e-01 0.14671917
## 174 1.407700e-04 0.08398557
## 175 5.020260e-01 0.40463309
## 176 1.407700e-04 0.08398557
## 177 1.407700e-04 0.08398557
## 178 1.060202e-01 0.14671917
## 179 1.060202e-01 0.14671917
## 180 1.060202e-01 0.14671917
## 181 5.020260e-01 0.40463309
## 182 5.020260e-01 0.40463309
## 183 5.020260e-01 0.40463309
## 184 1.060202e-01 0.14671917
## 185 1.060202e-01 0.14671917
## 186 1.060202e-01 0.14671917
## 187 1.407700e-04 0.08398557
## 188 5.020260e-01 0.40463309
## 189 5.020260e-01 0.40463309
## 190 1.060202e-01 0.14671917
## 191 1.060202e-01 0.14671917
## 192 1.060202e-01 0.14671917
```

```
## 193 5.020260e-01 0.40463309
## 194 1.060202e-01 0.14671917
## 195 5.020260e-01 0.40463309
## 196 5.020260e-01 0.40463309
## 197 1.407700e-04 0.08398557
## 198 5.020260e-01 0.40463309
## 199 2.935665e-01 0.15360951
## 200 1.060202e-01 0.14671917
## 201 1.411115e-01 0.14081538
## 202 1.411115e-01 0.14081538
## 203 5.020260e-01 0.40463309
## 204 1.060202e-01 0.14671917
## 205 1.407700e-04 0.08398557
## 206 1.060202e-01 0.14671917
## 207 5.020260e-01 0.40463309
## 208 1.060202e-01 0.14671917
## 209 1.060202e-01 0.14671917
## 210 1.407700e-04 0.08398557
## 211 1.407700e-04 0.08398557
## 212 1.060202e-01 0.14671917
## 213 1.407700e-04 0.08398557
## 214 5.020260e-01 0.40463309
## 215 1.411115e-01 0.14081538
## 216 1.060202e-01 0.14671917
## 217 2.935665e-01 0.15360951
## 218 4.752597e-01 0.08855878
## 219 1.411115e-01 0.14081538
## 220 5.020260e-01 0.40463309
## 221 1.060202e-01 0.14671917
## 222 1.407700e-04 0.08398557
## 223 1.407700e-04 0.08398557
## 224 1.060202e-01 0.14671917
## 225 1.060202e-01 0.14671917
## 226 1.407700e-04 0.08398557
## 227 1.407700e-04 0.08398557
## 228 1.411115e-01 0.14081538
## 229 1.407700e-04 0.08398557
## 230 1.060202e-01 0.14671917
## 231 1.407700e-04 0.08398557
## 232 5.020260e-01 0.40463309
## 233 5.020260e-01 0.40463309
## 234 1.060202e-01 0.14671917
## 235 1.407700e-04 0.08398557
## 236 1.060202e-01 0.14671917
## 237 1.407700e-04 0.08398557
## 238 1.060202e-01 0.14671917
## 239 1.060202e-01 0.14671917
## 240 5.020260e-01 0.40463309
## 241 1.060202e-01 0.14671917
## 242 1.407700e-04 0.08398557
## 243 1.407700e-04 0.08398557
## 244 5.020260e-01 0.40463309
## 245 1.407700e-04 0.08398557
## 246 5.020260e-01 0.40463309
## 247 1.060202e-01 0.14671917
## 248 1.407700e-04 0.08398557
## 249 5.020260e-01 0.40463309
## 250 1.407700e-04 0.08398557
## 251 4.752597e-01 0.08855878
## 252 1.407700e-04 0.08398557
## 253 1.060202e-01 0.14671917
## 254 1.407700e-04 0.08398557
## 255 1.060202e-01 0.14671917
## 256 7.319838e-01 0.07891129
## 257 1.060202e-01 0.14671917
## 258 1.060202e-01 0.14671917
## 259 1.060202e-01 0.14671917
## 260 1.411115e-01 0.14081538
## 261 1.060202e-01 0.14671917
## 262 7.319838e-01 0.07891129
## 263 1.407700e-04 0.08398557
## 264 7.319838e-01 0.07891129
## 265 1.060202e-01 0.14671917
## 266 1.060202e-01 0.14671917
## 267 1.060202e-01 0.14671917
## 268 5.020260e-01 0.40463309
## 269 4.752597e-01 0.08855878
## 270 1.407700e-04 0.08398557
## 271 1.407700e-04 0.08398557
## 272 1.407700e-04 0.08398557
## 273 5.020260e-01 0.40463309
## 
## $cos2
##             Dim 1       Dim 2
## 1   2.455272e-01 0.250807832
## 2   6.764496e-01 0.017170741
## 3   8.773127e-02 0.064622713
## 4   2.455272e-01 0.250807832
## 5   2.455272e-01 0.250807832
## 6   2.940101e-04 0.129479226
## 7   2.940101e-04 0.129479226
## 8   4.421547e-01 0.003153948
## 9   2.940101e-04 0.129479226
## 10  2.870900e-01 0.039487690
## 11  2.940101e-04 0.129479226
## 12  1.333221e-02 0.681230021
## 13  6.764496e-01 0.017170741
## 14  3.586500e-02 0.011927232
## 15  1.800749e-03 0.118482271
## 16  1.926358e-02 0.002284652
## 17  2.455272e-01 0.250807832
## 18  2.147173e-01 0.001828133
## 19  1.333221e-02 0.681230021
## 20  1.790094e-01 0.001557883
## 21  3.080335e-01 0.042883717
## 22  8.343163e-02 0.053050594
## 23  2.736902e-05 0.361034543
## 24  2.455272e-01 0.250807832
## 25  2.940101e-04 0.129479226
## 26  2.940101e-04 0.129479226
## 27  2.147173e-01 0.001828133
## 28  3.080335e-01 0.042883717
## 29  8.773127e-02 0.064622713
## 30  8.773127e-02 0.064622713
## 31  8.773127e-02 0.064622713
## 32  3.080335e-01 0.042883717
## 33  8.773127e-02 0.064622713
## 34  2.940101e-04 0.129479226
## 35  1.926358e-02 0.002284652
## 36  2.455272e-01 0.250807832
## 37  3.586500e-02 0.011927232
## 38  7.993722e-02 0.004064192
## 39  2.505615e-01 0.003223595
## 40  6.333400e-01 0.030771314
## 41  2.192935e-01 0.003912039
## 42  2.870900e-01 0.039487690
## 43  2.455272e-01 0.250807832
## 44  2.455272e-01 0.250807832
## 45  4.041015e-01 0.240418992
## 46  2.940101e-04 0.129479226
## 47  2.455272e-01 0.250807832
## 48  6.764496e-01 0.017170741
## 49  2.940101e-04 0.129479226
## 50  2.455272e-01 0.250807832
## 51  4.041015e-01 0.240418992
## 52  4.370060e-01 0.030805417
## 53  2.455272e-01 0.250807832
## 54  8.773127e-02 0.064622713
## 55  3.041959e-01 0.024206616
## 56  1.244293e-01 0.048059280
## 57  2.940101e-04 0.129479226
## 58  2.940101e-04 0.129479226
## 59  1.333221e-02 0.681230021
```

```
## 60  2.527476e-01 0.248202912
## 61  2.940101e-04 0.129479226
## 62  4.041015e-01 0.240418992
## 63  2.455272e-01 0.250807832
## 64  2.736902e-05 0.361034543
## 65  2.940101e-04 0.129479226
## 66  3.041959e-01 0.024206616
## 67  2.940101e-04 0.129479226
## 68  2.455272e-01 0.250807832
## 69  8.343163e-02 0.053050594
## 70  2.940101e-04 0.129479226
## 71  4.041015e-01 0.240418992
## 72  4.041015e-01 0.240418992
## 73  8.343163e-02 0.053050594
## 74  2.940101e-04 0.129479226
## 75  4.041015e-01 0.240418992
## 76  2.455272e-01 0.250807832
## 77  2.940101e-04 0.129479226
## 78  4.041015e-01 0.240418992
## 79  8.343163e-02 0.053050594
## 80  1.333221e-02 0.681230021
## 81  4.041015e-01 0.240418992
## 82  8.343163e-02 0.053050594
## 83  2.192427e-05 0.126514480
## 84  1.333221e-02 0.681230021
## 85  6.764496e-01 0.017170741
## 86  4.370060e-01 0.030805417
## 87  3.041959e-01 0.024206616
## 88  2.455272e-01 0.250807832
## 89  2.940101e-04 0.129479226
## 90  8.343163e-02 0.053050594
## 91  8.773127e-02 0.064622713
## 92  2.940101e-04 0.129479226
## 93  2.940101e-04 0.129479226
## 94  2.940101e-04 0.129479226
## 95  2.940101e-04 0.129479226
## 96  6.764496e-01 0.017170741
## 97  2.870900e-01 0.039487690
## 98  2.870900e-01 0.039487690
## 99  4.041015e-01 0.240418992
## 100 4.041015e-01 0.240418992
## 101 4.041015e-01 0.240418992
## 102 2.940101e-04 0.129479226
## 103 2.455272e-01 0.250807832
## 104 2.940101e-04 0.129479226
## 105 4.041015e-01 0.240418992
## 106 2.940101e-04 0.129479226
## 107 2.455272e-01 0.250807832
## 108 4.041015e-01 0.240418992
## 109 2.192427e-05 0.126514480
## 110 2.455272e-01 0.250807832
## 111 3.041959e-01 0.024206616
## 112 1.244293e-01 0.048059280
## 113 1.333221e-02 0.681230021
## 114 1.333221e-02 0.681230021
## 115 2.455272e-01 0.250807832
## 116 2.455272e-01 0.250807832
## 117 2.455272e-01 0.250807832
## 118 4.041015e-01 0.240418992
## 119 4.041015e-01 0.240418992
## 120 2.455272e-01 0.250807832
## 121 2.940101e-04 0.129479226
## 122 2.455272e-01 0.250807832
## 123 2.455272e-01 0.250807832
## 124 4.041015e-01 0.240418992
## 125 4.041015e-01 0.240418992
## 126 2.455272e-01 0.250807832
## 127 6.764496e-01 0.017170741
## 128 2.940101e-04 0.129479226
## 129 4.041015e-01 0.240418992
## 130 2.940101e-04 0.129479226
## 131 8.773127e-02 0.064622713
## 132 2.940101e-04 0.129479226
## 133 2.455272e-01 0.250807832
## 134 2.455272e-01 0.250807832
## 135 2.455272e-01 0.250807832
## 136 2.870900e-01 0.039487690
## 137 2.940101e-04 0.129479226
## 138 2.940101e-04 0.129479226
## 139 2.940101e-04 0.129479226
## 140 6.764496e-01 0.017170741
## 141 2.455272e-01 0.250807832
## 142 2.940101e-04 0.129479226
## 143 2.455272e-01 0.250807832
## 144 1.244293e-01 0.048059280
## 145 2.455272e-01 0.250807832
## 146 8.773127e-02 0.064622713
## 147 4.041015e-01 0.240418992
## 148 2.455272e-01 0.250807832
## 149 2.455272e-01 0.250807832
## 150 2.455272e-01 0.250807832
## 151 2.455272e-01 0.250807832
## 152 2.940101e-04 0.129479226
## 153 4.041015e-01 0.240418992
## 154 2.940101e-04 0.129479226
## 155 3.041959e-01 0.024206616
## 156 4.041015e-01 0.240418992
## 157 2.940101e-04 0.129479226
## 158 4.370060e-01 0.030805417
## 159 2.940101e-04 0.129479226
## 160 4.041015e-01 0.240418992
## 161 1.244293e-01 0.048059280
## 162 2.455272e-01 0.250807832
## 163 2.870900e-01 0.039487690
## 164 2.455272e-01 0.250807832
## 165 4.041015e-01 0.240418992
## 166 2.455272e-01 0.250807832
## 167 2.455272e-01 0.250807832
## 168 2.455272e-01 0.250807832
## 169 2.455272e-01 0.250807832
## 170 2.870900e-01 0.039487690
## 171 2.455272e-01 0.250807832
## 172 2.940101e-04 0.129479226
## 173 2.455272e-01 0.250807832
## 174 2.940101e-04 0.129479226
## 175 4.041015e-01 0.240418992
## 176 2.940101e-04 0.129479226
## 177 2.940101e-04 0.129479226
## 178 2.455272e-01 0.250807832
## 179 2.455272e-01 0.250807832
## 180 2.455272e-01 0.250807832
## 181 4.041015e-01 0.240418992
## 182 4.041015e-01 0.240418992
## 183 4.041015e-01 0.240418992
## 184 2.455272e-01 0.250807832
## 185 2.455272e-01 0.250807832
## 186 2.455272e-01 0.250807832
## 187 2.940101e-04 0.129479226
## 188 4.041015e-01 0.240418992
## 189 4.041015e-01 0.240418992
## 190 2.455272e-01 0.250807832
## 191 2.455272e-01 0.250807832
## 192 2.455272e-01 0.250807832
## 193 4.041015e-01 0.240418992
## 194 2.455272e-01 0.250807832
## 195 4.041015e-01 0.240418992
## 196 4.041015e-01 0.240418992
## 197 2.940101e-04 0.129479226
## 198 4.041015e-01 0.240418992
## 199 1.244293e-01 0.048059280
## 200 2.455272e-01 0.250807832
## 201 8.773127e-02 0.064622713
## 202 8.773127e-02 0.064622713
```

```
## 203 4.041015e-01 0.240418992
## 204 2.455272e-01 0.250807832
## 205 2.940101e-04 0.129479226
## 206 2.455272e-01 0.250807832
## 207 4.041015e-01 0.240418992
## 208 2.455272e-01 0.250807832
## 209 2.455272e-01 0.250807832
## 210 2.940101e-04 0.129479226
## 211 2.940101e-04 0.129479226
## 212 2.455272e-01 0.250807832
## 213 2.940101e-04 0.129479226
## 214 4.041015e-01 0.240418992
## 215 8.773127e-02 0.064622713
## 216 2.455272e-01 0.250807832
## 217 1.244293e-01 0.048059280
## 218 2.870900e-01 0.039487690
## 219 8.773127e-02 0.064622713
## 220 4.041015e-01 0.240418992
## 221 2.455272e-01 0.250807832
## 222 2.940101e-04 0.129479226
## 223 2.940101e-04 0.129479226
## 224 2.455272e-01 0.250807832
## 225 2.455272e-01 0.250807832
## 226 2.940101e-04 0.129479226
## 227 2.940101e-04 0.129479226
## 228 8.773127e-02 0.064622713
## 229 2.940101e-04 0.129479226
## 230 2.455272e-01 0.250807832
## 231 2.940101e-04 0.129479226
## 232 4.041015e-01 0.240418992
## 233 4.041015e-01 0.240418992
## 234 2.455272e-01 0.250807832
## 235 2.940101e-04 0.129479226
## 236 2.455272e-01 0.250807832
## 237 2.940101e-04 0.129479226
## 238 2.455272e-01 0.250807832
## 239 2.455272e-01 0.250807832
## 240 4.041015e-01 0.240418992
## 241 2.455272e-01 0.250807832
## 242 2.940101e-04 0.129479226
## 243 2.940101e-04 0.129479226
## 244 4.041015e-01 0.240418992
## 245 2.940101e-04 0.129479226
## 246 4.041015e-01 0.240418992
## 247 2.455272e-01 0.250807832
## 248 2.940101e-04 0.129479226
## 249 4.041015e-01 0.240418992
## 250 2.940101e-04 0.129479226
## 251 2.870900e-01 0.039487690
## 252 2.940101e-04 0.129479226
## 253 2.455272e-01 0.250807832
## 254 2.940101e-04 0.129479226
## 255 2.455272e-01 0.250807832
## 256 3.041959e-01 0.024206616
## 257 2.455272e-01 0.250807832
## 258 2.455272e-01 0.250807832
## 259 2.455272e-01 0.250807832
## 260 8.773127e-02 0.064622713
## 261 2.455272e-01 0.250807832
## 262 3.041959e-01 0.024206616
## 263 2.940101e-04 0.129479226
## 264 3.041959e-01 0.024206616
## 265 2.455272e-01 0.250807832
## 266 2.455272e-01 0.250807832
## 267 2.455272e-01 0.250807832
## 268 4.041015e-01 0.240418992
## 269 2.870900e-01 0.039487690
## 270 2.940101e-04 0.129479226
## 271 2.940101e-04 0.129479226
## 272 2.940101e-04 0.129479226
## 273 4.041015e-01 0.240418992
```

```r
print("wykresy ggplot2 w roznych pakietach koordynant")
```

```
## [1] "wykresy ggplot2 w roznych pakietach koordynant"
```

```r
print("ramka dla data frame with variable coordinates")
```

```
## [1] "ramka dla data frame with variable coordinates"
```

```r
mca_vars_df <- data.frame(res.mca$var$coord, Variable = rep(names(cats), cats))
head(mca_vars_df)
```

```
##                        Dim.1        Dim.2    Variable
## Indian           -0.08927978 -0.061246790 Nationality
## Indonesian       -0.04994931  2.005628036 Nationality
## Malaysian         0.85553323 -0.323228490 Nationality
## Pakistani         1.38789819 -0.020691367 Nationality
## Traditional food -0.17822336  0.001254865        Food
## Western Food      0.90299836 -0.006357981        Food
```

```r
print("koordynanty data frame with observation coordinates")
```

```
## [1] "koordynanty data frame with observation coordinates"
```

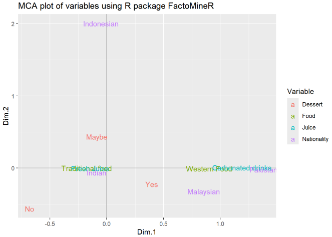
```r
mca_obs_df = data.frame(res.mca$ind$coord)
head(mca_obs_df)
```

```
##        Dim.1      Dim.2
## 1 -0.3311880  0.3347306
## 2  1.5834986 -0.2522868
## 3  0.3820864  0.3279269
## 4 -0.3311880  0.3347306
## 5 -0.3311880  0.3347306
## 6 -0.0120680 -0.2532528
```

```r
print( "wykres1 MCA: plot of variable categories")
```

```
## [1] "wykres1 MCA: plot of variable categories"
```

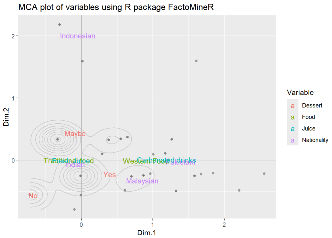
```r
ggplot(data=mca_vars_df, aes(x = Dim.1, y = Dim.2,
                             label = rownames(mca_vars_df)))+
  geom_hline(yintercept = 0, colour = "gray70")+
  geom_vline(xintercept = 0, colour = "gray70")+
  geom_text(aes(colour=Variable))+
  ggtitle("MCA plot of variables using R package FactoMineR")
```



```
print( "wykres2 MCA: plot of variable categories")
```

```
## [1] "wykres2 MCA: plot of variable categories"
```

```
ggplot(data = mca_obs_df, aes(x = Dim.1, y = Dim.2))+
  geom_hline(yintercept = 0, colour = "gray70")+
  geom_vline(xintercept = 0, colour = "gray70")+
  geom_point(colour = "gray50", alpha = 0.7)+
  geom_density2d(colour = "gray80")+
  geom_text(data = mca_vars_df, aes(x = Dim.1, y = Dim.2, label = rownames(mca_vars_df), colour = Variable))+ ggtitle("MCA p
lot of variables using R package FactoMineR")+
  scale_colour_discrete(name = "Variable")
```



```
#tworzenie nowych zmiennych"

nowa1<-matrix(0,nrow(X),1)
for(i in 1 : nrow(X)){
  if(X$Food[i]=="Traditional food" & X$Juice[i]=="Fresh Juice" & (X$Dessert[i]=="Maybe"|X$Dessert[i]=="No"|X$Dessert[i]=="Ye
s")){nowa1[i,1]<-"a"}
  if(X$Food[i]=="Western Food"& X$Juice[i]=="Carbonated drinks"&X$Dessert[i]=="Yes"){nowa1[i,1]<-"b"}
}
nowa1
```

```
##       [,1]
##  [1,] "a"
##  [2,] "b"
##  [3,] "0"
##  [4,] "a"
##  [5,] "a"
##  [6,] "a"
##  [7,] "a"
##  [8,] "b"
##  [9,] "a"
## [10,] "0"
## [11,] "a"
## [12,] "a"
## [13,] "b"
## [14,] "a"
## [15,] "a"
## [16,] "a"
## [17,] "a"
## [18,] "0"
## [19,] "a"
## [20,] "0"
## [21,] "0"
## [22,] "a"
## [23,] "a"
## [24,] "a"
## [25,] "a"
## [26,] "a"
## [27,] "0"
## [28,] "0"
## [29,] "0"
## [30,] "0"
## [31,] "0"
## [32,] "0"
## [33,] "0"
## [34,] "a"
## [35,] "a"
## [36,] "a"
## [37,] "a"
## [38,] "a"
## [39,] "0"
## [40,] "b"
## [41,] "0"
## [42,] "0"
## [43,] "a"
## [44,] "a"
## [45,] "a"
## [46,] "a"
## [47,] "a"
## [48,] "b"
## [49,] "a"
## [50,] "a"
## [51,] "a"
## [52,] "0"
## [53,] "a"
## [54,] "0"
## [55,] "0"
## [56,] "0"
## [57,] "a"
## [58,] "a"
## [59,] "a"
## [60,] "b"
## [61,] "a"
## [62,] "a"
## [63,] "a"
```

```
##  [64,] "a"
##  [65,] "a"
##  [66,] "0"
##  [67,] "a"
##  [68,] "a"
##  [69,] "a"
##  [70,] "a"
##  [71,] "a"
##  [72,] "a"
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##  [78,] "a"
##  [79,] "a"
##  [80,] "a"
##  [81,] "a"
##  [82,] "a"
##  [83,] "0"
##  [84,] "a"
##  [85,] "b"
##  [86,] "0"
##  [87,] "0"
##  [88,] "a"
##  [89,] "a"
##  [90,] "a"
##  [91,] "0"
##  [92,] "a"
##  [93,] "a"
##  [94,] "a"
##  [95,] "a"
##  [96,] "b"
##  [97,] "0"
##  [98,] "0"
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## [126,] "a"
## [127,] "b"
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## [136,] "0"
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## [139,] "a"
## [140,] "b"
## [141,] "a"
## [142,] "a"
## [143,] "a"
## [144,] "0"
## [145,] "a"
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## [169,] "a"
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## [196,] "a"
## [197,] "a"
## [198,] "a"
## [199,] "0"
## [200,] "a"
## [201,] "0"
## [202,] "0"
## [203,] "a"
## [204,] "a"
## [205,] "a"
```

```
## [206,] "a"
## [207,] "a"
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## [211,] "a"
## [212,] "a"
## [213,] "a"
## [214,] "a"
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## [216,] "a"
## [217,] "0"
## [218,] "0"
## [219,] "0"
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## [227,] "a"
## [228,] "0"
## [229,] "a"
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## [250,] "a"
## [251,] "0"
## [252,] "a"
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## [267,] "a"
## [268,] "a"
## [269,] "0"
## [270,] "a"
## [271,] "a"
## [272,] "a"
## [273,] "a"
```

```
gr1<-data.frame(nowa1)
colnames(gr1)<-c("grupy")

#nowe dane
WYN<-data.frame(X,nowa1)
head(WYN,20)
```

```
##    Nationality             Food             Juice Dessert nowa1
## 1       Indian Traditional food       Fresh Juice   Maybe     a
## 2       Indian     Western Food Carbonated drinks     Yes     b
## 3       Indian     Western Food       Fresh Juice   Maybe     0
## 4       Indian Traditional food       Fresh Juice   Maybe     a
## 5       Indian Traditional food       Fresh Juice   Maybe     a
## 6       Indian Traditional food       Fresh Juice     Yes     a
## 7       Indian Traditional food       Fresh Juice     Yes     a
## 8    Pakistani     Western Food Carbonated drinks     Yes     b
## 9       Indian Traditional food       Fresh Juice     Yes     a
## 10      Indian     Western Food       Fresh Juice     Yes     0
## 11      Indian Traditional food       Fresh Juice     Yes     a
## 12  Indonesian Traditional food       Fresh Juice   Maybe     a
## 13      Indian     Western Food Carbonated drinks     Yes     b
## 14   Pakistani Traditional food       Fresh Juice   Maybe     a
## 15   Malaysian Traditional food       Fresh Juice      No     a
## 16   Malaysian Traditional food       Fresh Juice   Maybe     a
## 17      Indian Traditional food       Fresh Juice   Maybe     a
## 18   Malaysian Traditional food Carbonated drinks   Maybe     0
## 19  Indonesian Traditional food       Fresh Juice   Maybe     a
## 20   Malaysian     Western Food       Fresh Juice   Maybe     0
```

```
table(gr1)
```

```
## grupy
##   0   a   b
##  52 211  10
```

```
#przypadki nieujete w grupowaniu
subset(WYN,WYN$nowa1==0)
```

```
##     Nationality             Food             Juice Dessert nowa1
## 3        Indian     Western Food       Fresh Juice   Maybe     0
## 10       Indian     Western Food       Fresh Juice     Yes     0
## 18    Malaysian Traditional food Carbonated drinks   Maybe     0
## 20    Malaysian     Western Food       Fresh Juice   Maybe     0
## 21    Malaysian     Western Food       Fresh Juice     Yes     0
## 27    Malaysian Traditional food Carbonated drinks   Maybe     0
## 28    Malaysian     Western Food       Fresh Juice     Yes     0
## 29       Indian     Western Food       Fresh Juice   Maybe     0
## 30       Indian     Western Food       Fresh Juice   Maybe     0
## 31       Indian     Western Food       Fresh Juice   Maybe     0
## 32    Malaysian     Western Food       Fresh Juice     Yes     0
## 33       Indian     Western Food       Fresh Juice   Maybe     0
## 39    Pakistani Traditional food Carbonated drinks     Yes     0
## 41    Pakistani     Western Food       Fresh Juice     Yes     0
## 42       Indian     Western Food       Fresh Juice     Yes     0
## 52       Indian     Western Food Carbonated drinks   Maybe     0
## 54       Indian     Western Food       Fresh Juice   Maybe     0
## 55       Indian Traditional food Carbonated drinks     Yes     0
## 56       Indian Traditional food Carbonated drinks   Maybe     0
## 66       Indian Traditional food Carbonated drinks     Yes     0
## 83       Indian     Western Food       Fresh Juice      No     0
## 86       Indian     Western Food Carbonated drinks   Maybe     0
## 87       Indian Traditional food Carbonated drinks     Yes     0
## 91       Indian     Western Food       Fresh Juice   Maybe     0
## 97       Indian     Western Food       Fresh Juice     Yes     0
## 98       Indian     Western Food       Fresh Juice     Yes     0
## 109      Indian     Western Food       Fresh Juice      No     0
## 111      Indian Traditional food Carbonated drinks     Yes     0
## 112      Indian Traditional food Carbonated drinks   Maybe     0
## 131      Indian     Western Food       Fresh Juice   Maybe     0
## 136      Indian     Western Food       Fresh Juice     Yes     0
## 144      Indian Traditional food Carbonated drinks   Maybe     0
## 146      Indian     Western Food       Fresh Juice   Maybe     0
## 155      Indian Traditional food Carbonated drinks     Yes     0
## 158      Indian     Western Food Carbonated drinks   Maybe     0
## 161      Indian Traditional food Carbonated drinks   Maybe     0
## 163      Indian     Western Food       Fresh Juice     Yes     0
## 170      Indian     Western Food       Fresh Juice     Yes     0
## 199      Indian Traditional food Carbonated drinks   Maybe     0
## 201      Indian     Western Food       Fresh Juice   Maybe     0
## 202      Indian     Western Food       Fresh Juice   Maybe     0
## 215      Indian     Western Food       Fresh Juice   Maybe     0
## 217      Indian Traditional food Carbonated drinks   Maybe     0
## 218      Indian     Western Food       Fresh Juice     Yes     0
## 219      Indian     Western Food       Fresh Juice   Maybe     0
## 228      Indian     Western Food       Fresh Juice   Maybe     0
## 251      Indian     Western Food       Fresh Juice     Yes     0
## 256      Indian Traditional food Carbonated drinks     Yes     0
## 260      Indian     Western Food       Fresh Juice   Maybe     0
## 262      Indian Traditional food Carbonated drinks     Yes     0
## 264      Indian Traditional food Carbonated drinks     Yes     0
## 269      Indian     Western Food       Fresh Juice     Yes     0
```

```
pop <- cbind(pop, C=WYN$nowa1)
write.table(pop, file="lab3result.csv",append=TRUE,sep=";",row.names=TRUE,col.names=TRUE,dec = ",")
```

```
## Warning in write.table(pop, file = "lab3result.csv", append = TRUE, sep = ";",
## : dołączanie nazw kolumn do pliku
```